

NAKHON PHANOM, Thailand

WMO: 483570

Lat: 17.410N

Lon: 104.776E

Elev: 148

StdP: 99.56

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 13.6	(c) 15.0	(d) 7.1	(e) 6.4	(f) 18.2	(g) 8.9	(h) 7.2	(i) 18.4	(j) 4.9	(k) 23.7	(l) 4.4	(m) 23.9	(n) 0.1	(o) 90	(p) 0.288

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 9.3	(c) 36.9	(d) 25.8	(e) 35.7	(f) 25.6	(g) 34.6	(h) 25.6	(i) 27.9	(j) 32.9	(k) 27.4	(l) 32.1	(m) 27.1	(n) 31.6	(o) 1.1	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.7	22.7	30.2	26.3	22.1	29.5	26.0	21.8	29.1	90.5	33.0	88.2	32.2	86.6	31.8	31.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
4.0	3.4	3.0	DB	10.3	38.6	1.8	1.3	9.0	39.5	8.0	40.3	7.0	41.0	5.7	41.9	
			WB	9.0	29.1	1.7	1.2	7.7	30.0	6.7	30.7	5.8	31.3	4.5	32.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.9	22.9	25.1	27.8	29.6	29.3	28.6	27.8	27.7	27.8	27.4	25.6	23.2
	DBStd	3.03	2.59	3.10	2.99	2.35	1.83	1.54	1.59	1.37	1.35	1.51	2.10	2.65
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	4	2	1	0	0	0	0	0	0	0	0	0	2
	CDD10.0	6170	401	423	553	588	597	557	552	549	534	539	468	408
	CDD18.3	3133	144	191	295	338	339	307	293	291	284	281	218	152
	CDH23.3	32718	1191	1811	3303	4193	4002	3435	2983	2904	2869	2816	1963	1248
	CDH26.7	13315	394	800	1633	2180	1818	1342	981	909	998	1106	748	406
Wind	WSAvg	0.7	1.0	0.9	0.9	0.7	0.5	0.4	0.3	0.4	0.4	0.6	0.8	1.0
Precipitation	PrecAvg	2422	2	12	51	89	259	564	432	660	296	62	5	2
	PrecMax	2837	10	49	118	204	443	1033	771	1233	587	291	45	29
	PrecMin	1820	0	0	6	9	164	288	271	274	97	7	0	0
	PrecStd	347	3	15	37	56	69	225	125	263	133	69	11	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.4	36.0	38.4	38.9	37.3	35.1	34.4	34.1	34.0	33.9	34.0	33.0
		MCWB	22.5	23.0	24.5	26.4	27.1	27.1	27.0	27.1	26.2	25.2	24.6	23.4
	2%	DB	31.7	34.5	36.4	37.3	35.7	34.0	33.2	32.8	33.1	33.1	32.7	31.6
		MCWB	22.1	23.4	24.8	26.3	26.7	26.8	26.8	26.8	26.0	25.0	24.1	22.8
	5%	DB	30.4	33.0	35.1	35.9	34.5	33.0	32.1	31.8	32.3	32.4	31.8	30.4
		MCWB	21.5	23.2	24.7	26.0	26.5	26.6	26.7	26.6	25.8	24.7	23.6	22.1
	10%	DB	29.1	31.4	33.7	34.6	33.2	32.0	31.0	30.8	31.4	31.6	30.7	29.1
		MCWB	21.0	22.7	24.4	25.7	26.3	26.6	26.4	26.3	25.7	24.4	23.0	21.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.7	26.0	26.9	28.7	29.5	28.0	27.7	27.7	27.5	26.9	25.8	24.9
		MCDB	30.3	32.5	34.5	35.6	34.4	32.8	32.1	31.8	31.6	31.5	31.1	30.3
	2%	WB	23.5	25.0	26.1	27.6	28.1	27.5	27.2	27.2	27.0	26.2	25.3	23.7
		MCDB	28.8	31.5	33.5	34.1	32.9	31.9	31.3	31.1	31.1	30.7	30.3	29.5
	5%	WB	22.5	24.2	25.5	26.9	27.2	27.1	27.0	26.9	26.7	25.7	24.7	22.9
		MCDB	28.2	30.3	32.6	33.3	32.2	31.4	30.8	30.5	30.4	30.0	29.5	28.9
	10%	WB	21.7	23.5	25.0	26.3	26.8	26.8	26.6	26.6	26.3	25.3	24.0	22.1
		MCDB	27.7	29.4	31.7	32.5	31.5	30.7	30.1	29.8	29.8	29.4	28.7	28.0
Mean Daily Temperature Range	5% DB	MDBR	11.6	11.4	10.5	9.3	7.6	6.3	5.6	5.7	6.5	8.2	9.8	11.0
		MCDBR	12.5	13.0	11.7	10.6	9.1	7.7	7.3	7.3	8.0	9.0	10.3	11.4
		MCWBR	4.5	4.0	3.4	3.2	2.9	2.4	2.4	2.4	2.6	2.7	3.2	4.0
	5% WB	MCDBR	11.3	11.2	10.6	9.5	8.0	6.9	6.5	6.7	7.0	7.8	9.0	10.6
		MCWBR	4.7	4.5	3.7	3.6	3.1	2.6	2.5	2.6	2.6	2.7	3.2	4.2
Clear-Sky Solar Irradiance	taub		0.452	0.559	0.663	0.633	0.512	0.475	0.462	0.474	0.505	0.512	0.454	0.455
	taud		2.173	1.894	1.694	1.800	2.153	2.281	2.312	2.266	2.170	2.115	2.250	2.185
	Ebn at Noon		827	755	693	718	799	822	833	830	806	786	821	811
	Edh at Noon		142	197	247	223	155	135	131	138	151	155	131	137
All-Sky Solar Radiation	RadAvg		4.67	5.00	4.99	5.33	5.09	4.66	4.30	4.30	4.52	4.73	4.66	4.41
	RadStd		0.28	0.36	0.21	0.29	0.40	0.45	0.53	0.37	0.40	0.41	0.21	0.17

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		+0.30	N/A	N/A	N/A	+0.20	+0.15	N/A	N/A	N/A	N/A

Nomenclature: See separate page